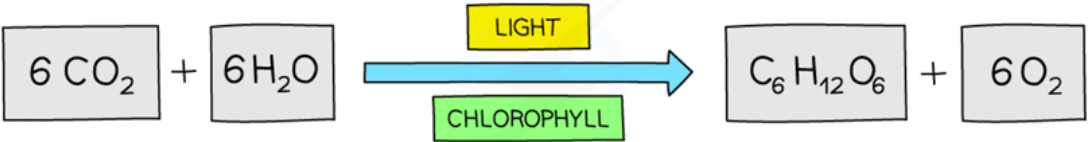
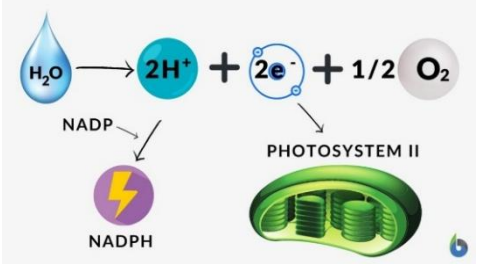
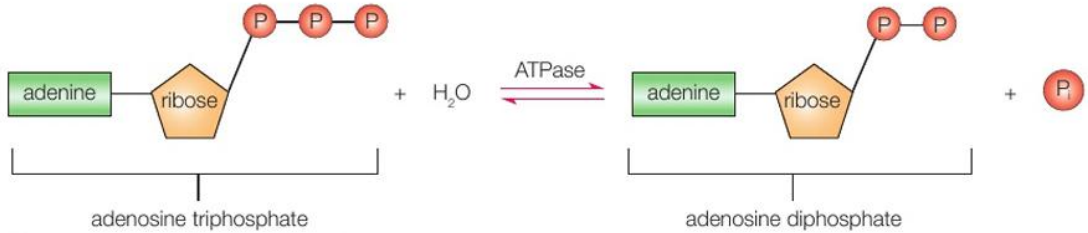


Unit 4: Energy, Environment, Microbiology and Immunity

Topic 5 – Energy Flow, Ecosystems and the Environment

Candidates will be assessed on their ability to:

5.1	understand the overall reaction of photosynthesis as requiring energy from light to split apart the strong bonds in water molecules, storing the hydrogen in a fuel (glucose) by combining it with carbon dioxide and releasing oxygen into the atmosphere
	Overall reaction of photosynthesis:  Photolysis of water: - Occurs in thylakoid membranes of chloroplasts - Covalent bond of a water molecule is broken - Supply hydrogen ions to reduce carbon dioxide which is then stored as glucose - Supply electrons for photosystem II (680nm) - Oxygen is produced as a waste product  Light Energy: - Light energy cannot be used directly as ATP is the source of energy 1. Light energy is absorbed by photosystems 2. This excites electrons 3. These electrons are passed along a series of electron carriers 4. Therefore, releasing energy (NOT ATP) to phosphorylate ADP into ATP (cyclic)] 5. Phosphorylation of ADP via the proton gradient to form ATP (non-cyclic) → This is because it also produces hydrogen ions which can be used in chemiosmosis
5.2	understand how photophosphorylation of ADP requires energy and that hydrolysis of ATP provides an immediate supply of energy for biological processes
	 Forward Reaction: - Hydrolysis - Releases energy (34kJ) Backward Reaction: - Condensation - Needs energy (34kJ) *Both require ATPase enzyme as catalyst.

5.3

understand the light-dependent reactions of photosynthesis, including how light energy is trapped by exciting electrons in chlorophyll and the role of these electrons in generating ATP, reducing NADP in cyclic and non-cyclic photophosphorylation and producing oxygen through photolysis of water

Light-Dependent Reaction:

- This occurs in thylakoid membrane
- ATP is made by photophosphorylation in light-dependent reaction.

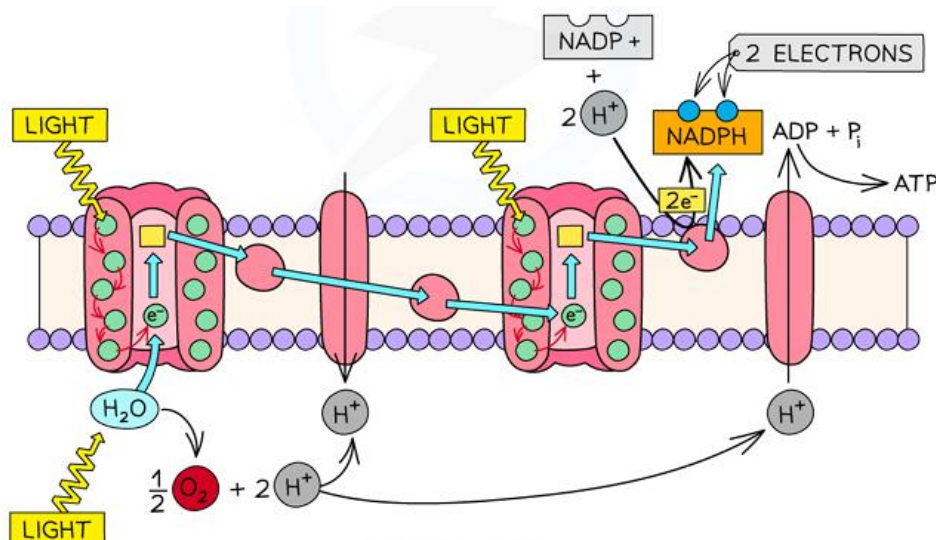
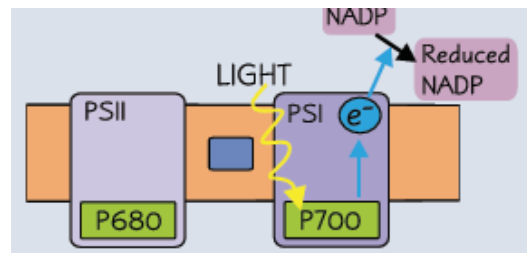
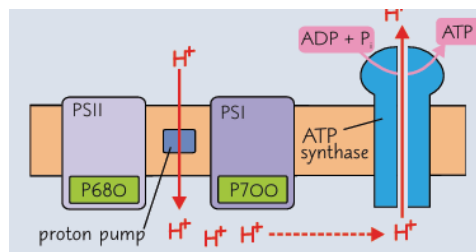
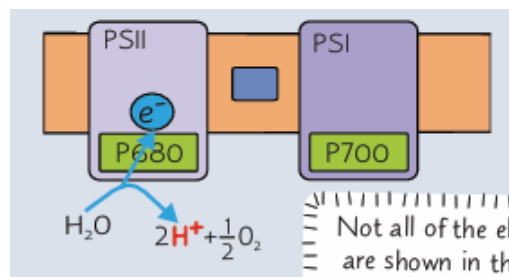
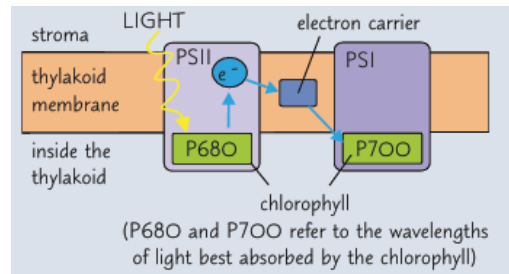
(2) types of Photophosphorylation

1. Non-Cyclic Photophosphorylation
2. Cyclic Photophosphorylation

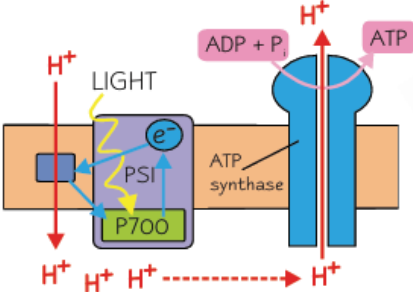
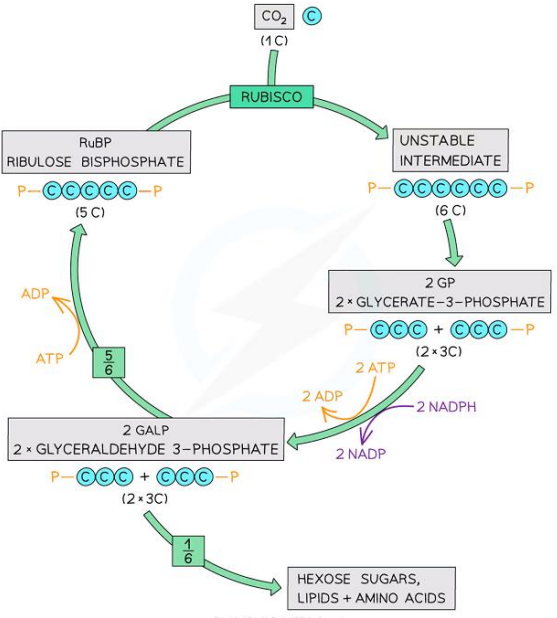
1. Non-Cyclic Photophosphorylation

1. PSII absorbs light energy to **excite** electron to move to higher energy level.
2. The electron in PSII is replaced by electron from **photolysis** of water.
3. The excited electron loses energy as it moves along the **electron transport chain**.

- This **energy** (NOT ATP) transports protons into thylakoid intermembrane space to form proton gradient and this produces **ATP** (chemiosmosis).
- 4. Light energy is absorbed by PSI, which excites the electron again to even higher energy level.
- The electron is finally transferred to NADP along with hydrogen ion from the stroma to form **reduced NADP**.

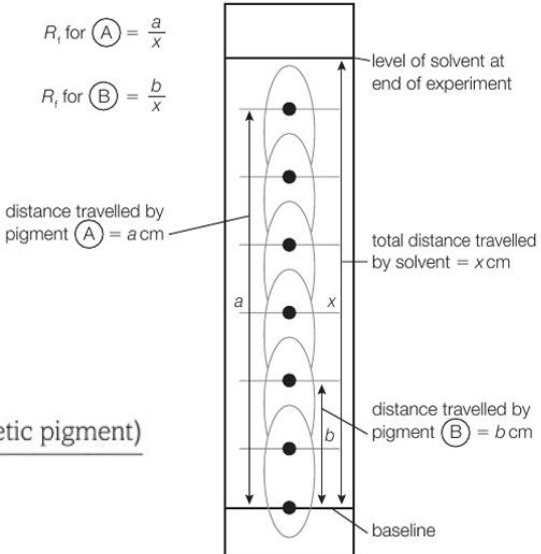
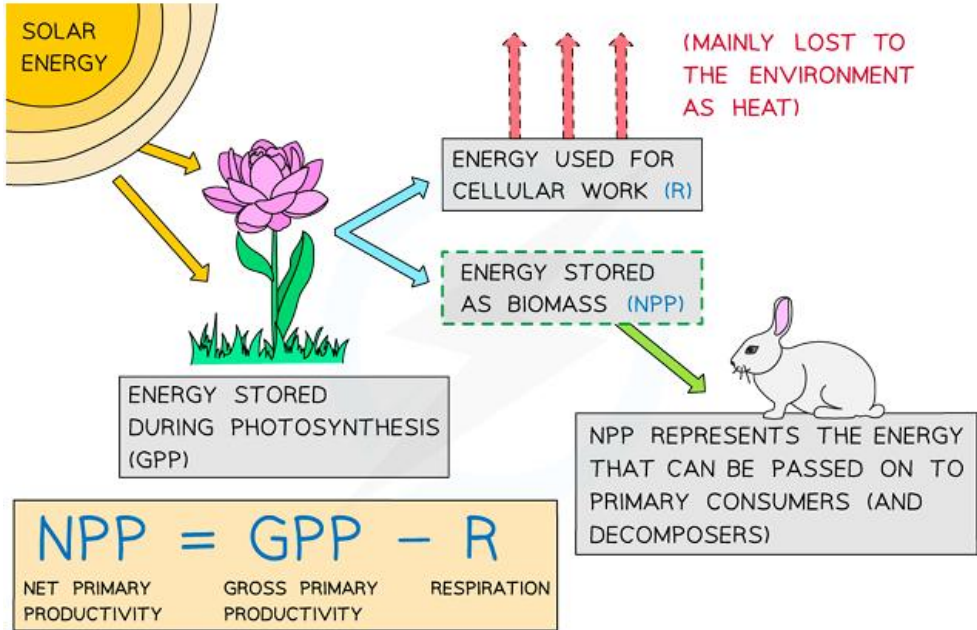


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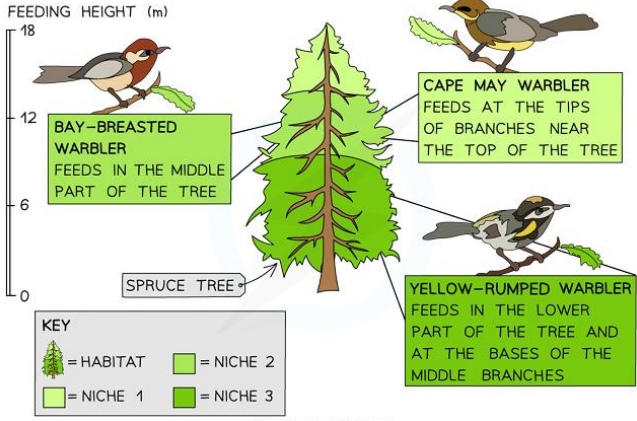
	2.Cyclic Photophosphorylation → Only uses PSI. → It is 'cyclic' as the electrons from chlorophyll molecule are not passed onto NADP, but are passed back to PSI via electron carrier. → This process does not produce any reduced NADP or O₂, only produces small amounts of ATP.  The diagram shows a cross-section of a thylakoid membrane. Light energy is absorbed by Photosystem I (PSI), which contains the P700 chlorophyll. Electrons (e⁻) are excited and pass through an electron carrier to ATP synthase. This process pumps protons (H⁺) from the stroma into the thylakoid space. The resulting proton gradient drives the synthesis of ATP from ADP and inorganic phosphate (P_i) as protons flow back through ATP synthase. The electrons return to PSI, completing the cycle. *Both reduced NADP and ATP enter the light-independent reaction: → ATP is required to provide energy to convert GP to GALP → Reduced NADP is used to reduce GP to GALP
5.4	(i) understand the light-independent reactions as reduction of carbon dioxide using the products of the light-dependent reactions (carbon fixation in the Calvin cycle, the role of GP, GALP, RuBP and RUBISCO) (ii) know that the products are simple sugars that are used by plants, animals and other organisms in respiration and the synthesis of new biological molecules (polysaccharides, amino acids, proteins, lipids and nucleic acids)
i	Light-Independent Reaction (Calvin Cycle): → Calvin cycle happens inside the stroma of chloroplast. (4) Steps of Calvin cycle: Carbon Fixation: RUBP (Ribulose Biphosphate) (5C compound) fixes CO₂ to produce (2) GP (Glycerate-3-Phosphate) (3C compound). → This is catalyzed by RUBISCO (Ribulose Biphosphate Carboxylase). Reduction: GP is reduced into GALP (Glyceraldehyde 3-Phosphate) using hydrogen from reduced NADP and energy from hydrolysis of ATP. Regeneration: GALP is converted back into RUBP. → This uses energy from hydrolysis of ATP. Glucose Formation: Every 6th turn of the Calvin cycle, (1) molecule of glucose is produced from GALP.  The diagram illustrates the Calvin cycle as a circular pathway. It begins with RuBP (Ribulose Biphosphate, 5C) combining with CO₂ (1C) in a reaction catalyzed by RUBISCO to form an unstable intermediate (6C). This intermediate splits into 2 GP (Glycerate-3-Phosphate, 3C). GP is then reduced to 2 GALP (Glyceraldehyde 3-Phosphate, 3C) using 2 ATP (which become 2 ADP) and 2 reduced NADP (which become 2 NADP). One GALP molecule (1/6) is used for glucose formation, while the remaining 5/6 are used to regenerate 2 RuBP (5C) molecules, completing the cycle. The final products are hexose sugars, lipids, and amino acids.
ii	Products of Light Independent Reactions: - GALP converted glucose every 6th turn of Calvin cycle Cellulose: - Beta-glucose arranged in reverse structure to form 1,4-glycosidic bonds Lipid: - Glucose is used to make glycerol and fatty acids

	- Glycerol and fatty acids joined by ester bonds in a condensation reaction, forming lipids DNA: - Glucose is converted into deoxyribose - Then, nitrogen from nitrates is added to glucose, forming a nitrogenous base and phosphate from phosphate ions are added to form DNA nucleotide Proteins: - Amino acids made from glucose and nitrates taken up from the soil - Amino acids joined by peptide bonds - Glucose respired to produce ATP for protein synthesis Biomass: - Glucose and fructose are used to produce sucrose - Amino acids made from glucose and nitrates taken up from the soil - Sucrose and amino acids are transported to the roots through the phloem - They are used to synthesize organic molecules such as starch, cellulose and proteins - Calcium ions are used to make calcium pectate for cell walls
5.5	understand the structure of chloroplasts in relation to their role in photosynthesis
	(ii) Compare and contrast the structure of the outer membrane of a chloroplast with that of a thylakoid membrane. www.RocketRevise.com (4) ...Both consist of phospholipid bilayer..... Both have proteins..... Thylakoid membrane has ATP synthase but outer membrane does not..... Thylakoid membrane has chlorophyll / PSI / PSII while outer membrane does not..... Thylakoid membrane has electron carriers but outer membrane does not.....

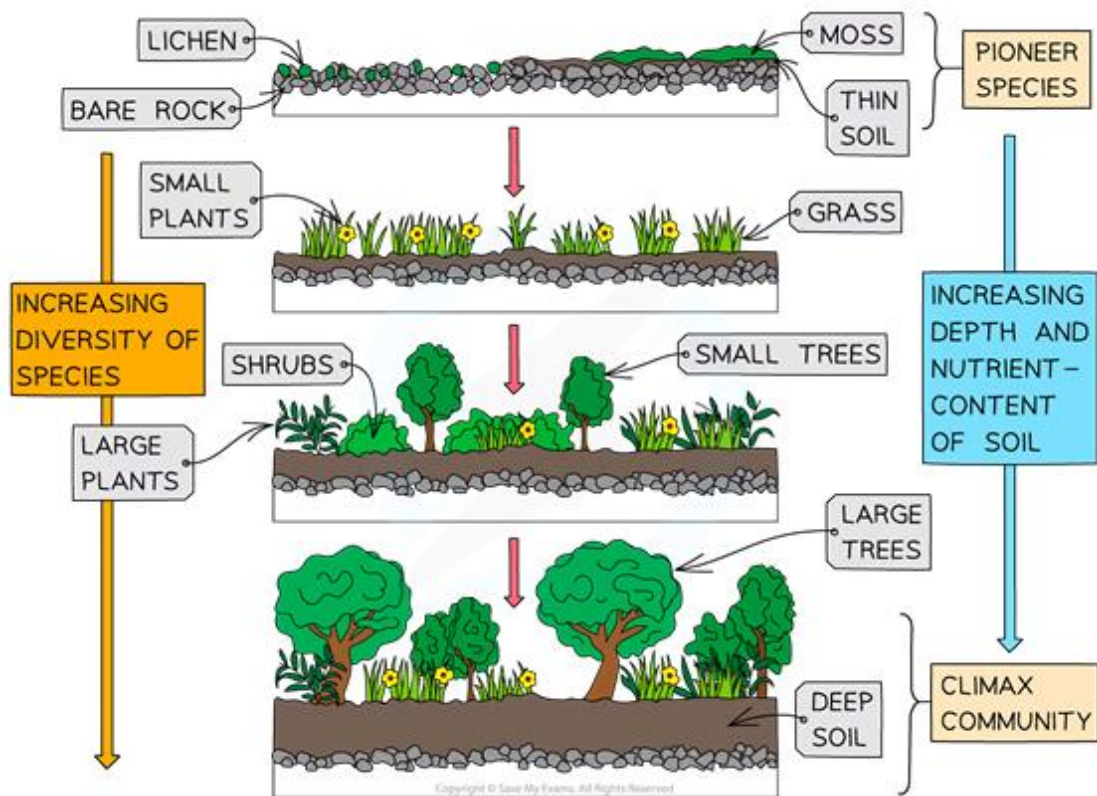
	Structures with Functions: - Chloroplast envelope: space between inner and outer membrane - Grana: stacks of thylakoids - Thylakoids: → double membrane; contains photosystems → light dependent reaction occurs in thylakoid membrane → this is where chlorophyll captures light and uses it to produce ATP - Lamellae: connects grana - Stroma: → gel-like fluid that contains all the enzymes needed for photosynthesis → light independent reaction occurs (Calvin cycle) - DNA - Ribosomes
5.6	understand what is meant by the terms <i>absorption spectrum</i> and <i>action spectrum</i>
	Absorption Spectrum: Rate of absorption of light at different wavelengths by different pigments Action Spectra: Rate of photosynthesis with wavelength of light *To determine color of the pigment based on the wavelength of light absorbed: - Chlorophyll A: Do not absorb yellow-green so its color is yellow-green • So, the color of wavelength it does not absorb is the color of pigment because that unabsorbed light is reflected back to give its color CHLOROPHYLLS AND CAROTENOIDS BOTH ABSORB WAVELENGTHS IN THE BLUE-VIOLET REGION OF THE LIGHT SPECTRUM CHLOROPHYLLS ALSO ABSORB WAVELENGTHS IN THE RED REGION OF THE LIGHT SPECTRUM, WHEREAS CAROTENOIDS DO NOT AMOUNT OF LIGHT ABSORBED WAVELENGTH OF LIGHT (nm) PHOTOSYNTHETIC RATE / ABSORPTION RATE WAVELENGTH (nm) KEY = ACTION SPECTRUM = CHLOROPHYLL A = β-CAROTENE = CHLOROPHYLL B = ABSORPTION SPECTRA
5.7	understand that chloroplast pigments can be separated using chromatography and the pigments identified using R _f values

	How a chromatogram can be produced: - Pigment dissolved in a solvent (ethanol/water) - Pigment will then be loaded onto the filter paper - Filter paper will then be dipped into the same solvent - This is left to run until the leading edge is near the end of filter paper To determine R_f value: $R_f \text{ value} = \frac{\text{distance travelled by solute (photosynthetic pigment)}}{\text{distance travelled by solvent}}$  $R_f \text{ for (A)} = \frac{a}{x}$ $R_f \text{ for (B)} = \frac{b}{x}$ distance travelled by pigment (A) = a cm total distance travelled by solvent = x cm distance travelled by pigment (B) = b cm level of solvent at end of experiment baseline
5.8	CORE PRACTICAL 10 Investigate the effects of light intensity, light wavelength, temperature and availability of carbon dioxide on the rate of photosynthesis using a suitable aquatic plant.
5.9	(i) understand the relationship between gross primary productivity (GPP), net primary productivity (NPP) and plant respiration (R) (ii) be able to calculate net primary productivity
i	Gross Primary Productivity: the rate at which light energy is converted into carbohydrates during photosynthesis Net Primary Productivity: the rate at which light energy is stored in plant biomass R (Respiratory Losses): the carbohydrates used up by the plant that are not stored in plant biomass →Units for ALL: $\text{kJ m}^{-2} \text{ yr}^{-1}$
ii	 SOLAR ENERGY → ENERGY STORED DURING PHOTOSYNTHESIS (GPP) ENERGY USED FOR CELLULAR WORK (R) → (MAINLY LOST TO THE ENVIRONMENT AS HEAT) ENERGY STORED AS BIOMASS (NPP) → NPP REPRESENTS THE ENERGY THAT CAN BE PASSED ON TO PRIMARY CONSUMERS (AND DECOMPOSERS) Formula: $NPP = GPP - R$ NET PRIMARY PRODUCTIVITY GROSS PRIMARY PRODUCTIVITY RESPIRATION
5.10	know how to calculate the efficiency of biomass and energy transfers between trophic levels

	Between Sun and Producer (Plant): - Not all wavelengths of light can be absorbed by the plant - Green light is reflected off the leaf - Not all the light falls on the leaves - Passes through leaf Between Consumers: - Inedible parts in organisms - Energy lost in respiration as heat - Egestion of indigestible parts of organism - Decomposers feed on the dead organisms by aerobic respiration Why trophic level is limited: - Energy is lost between trophic levels - So not enough energy for another trophic level Percentage Difference = $\frac{(\text{Final Value} - \text{Initial Value})}{\text{Initial Value}} \times 100$ Mean Percentage Difference between A and B = $\frac{\frac{A - B}{A + B}}{2} \times 100$ Percentage Efficiency of Energy Transfer = $\frac{\text{Energy available in one trophic level}}{\text{Energy available in previous trophic level}} \times 100$
5.11	understand what is meant by the terms <i>population</i> , <i>community</i> , <i>habitat</i> and <i>ecosystem</i>
	Population: organisms of one species found in a particular area Community: a group of organisms of different species interacting in a particular area Habitat: a place where organisms live Ecosystem: all organisms interacting with each other and the abiotic factors Biodiversity: variety of different species in a habitat and the genetic variation (different alleles) within those species Biomass: how much organic matter present in organisms
5.12	understand that the numbers and distribution of organisms in a habitat are controlled by biotic and abiotic factors
	Abiotic Factors: 1.Light: → Influence photosynthesis rate → Affect plant growth rate 2.Temperature: → Affect enzyme activity in animals and plants (RUBISCO) → Higher temperature results in higher kinetic energy of substrates so form more enzyme-substrate complexes 3.Water Availability: → For photosynthesis (light-dependent reaction) and for drinking for animals 4.Oxygen Availability: → For aerobic respiration;

	→ In soil, active transport process in roots for uptake of minerals → In ocean, for aquatic animals 5.Edaphic Factors (Soil Structure and Mineral Content) → Loam is the ideal soil as leaching of minerals is less likely and easy to drain water Biotic Factors: 1.Predation: → Population of both predator and prey will oscillate in a repeating cycle if other factors (such as parasitism) are not affected 2.Territory: → Purpose is to ensure that breeding pair has sufficient resources to raise their offspring safely 3.Parasitism and Disease: → If a species is sick, other competing species will outgrow if they are resistant → The whole population can be wiped out by the parasitism itself or due to the competition by other species with resistant genes 4.Finding a Mate: → Level of breeding depends on available mate founds. → If mates of same species are not found, hybridization may occur.
5.13	understand how the concept of niche accounts for the distribution and abundance of organisms in a habitat
	Niche: role of a living organism in its habitat  Adaptation to niche: - Anatomical: adaptations involving the form and structure of an organism - Physiological: adaptations involving the way the body of the organism works, including differences in biochemical pathways or enzymes - Behavioral: adaptation involving programmed or instinctive behavior making organisms better adapted for survival
5.14	CORE PRACTICAL 11 Carry out a study of the ecology of a habitat, such as using quadrats and transects to determine the distribution and abundance of organisms, and measuring abiotic factors appropriate to the habitat.
5.15	understand the stages of succession from colonization to the formation of a climax community

Succession: a sequence of changes in a community over a period of time
Climax Community: final stage of community that is self-sustaining and stable



Mechanism of Succession:

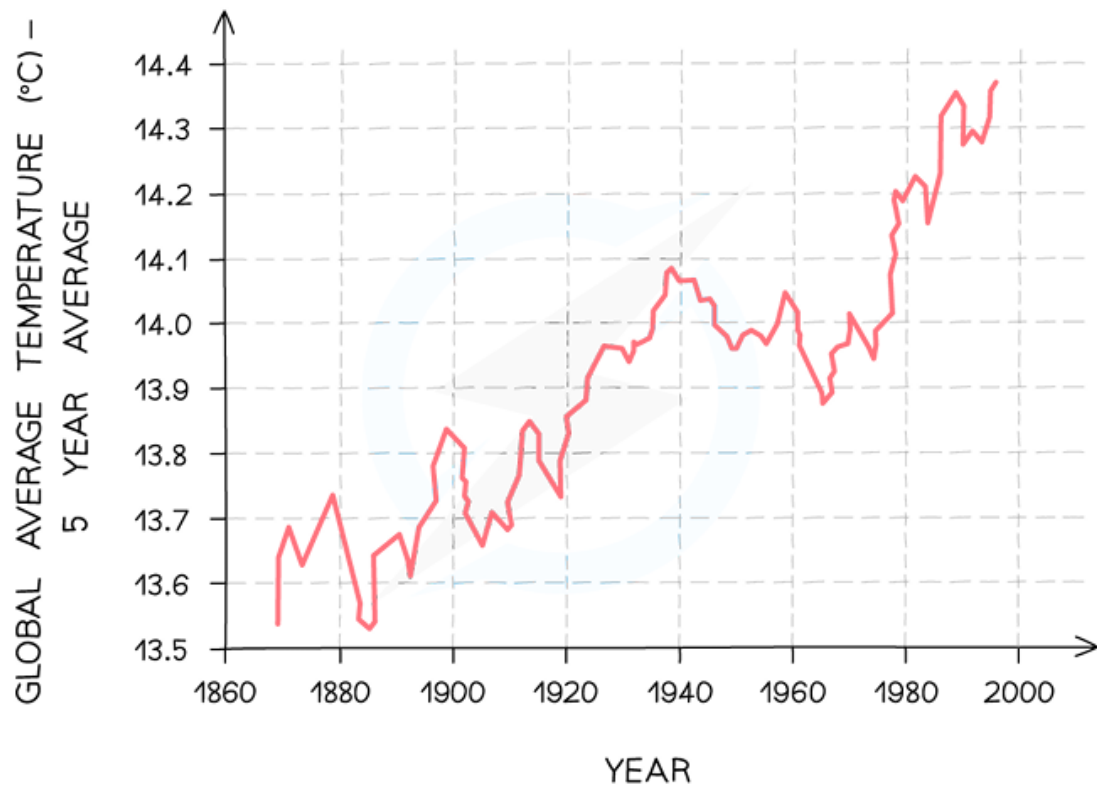
1. **Colonization: Pioneer species** such as algae and mosses produce enzymes and acids that digest the surface of bare rocks, releasing organic matter and forming a basic soil layer called **humus**.
2. This allows growth for short plants such as grass and shrubs, which forms more humus when they die and decay, which adds the **soil depth**.
 → This improves soil conditions such as soil texture, soil moisture, water and mineral availability.
3. The process continues with different plant species for over years until **climax community** is reached.

Characteristics of Pioneer Species:

- Need to withstand harsh environments
- Need to be able to withstand lack of shade
- Have a low requirement for minerals
- Have fast life-cycles and wide dispersal mechanisms

Why Succession occurs in Stages:

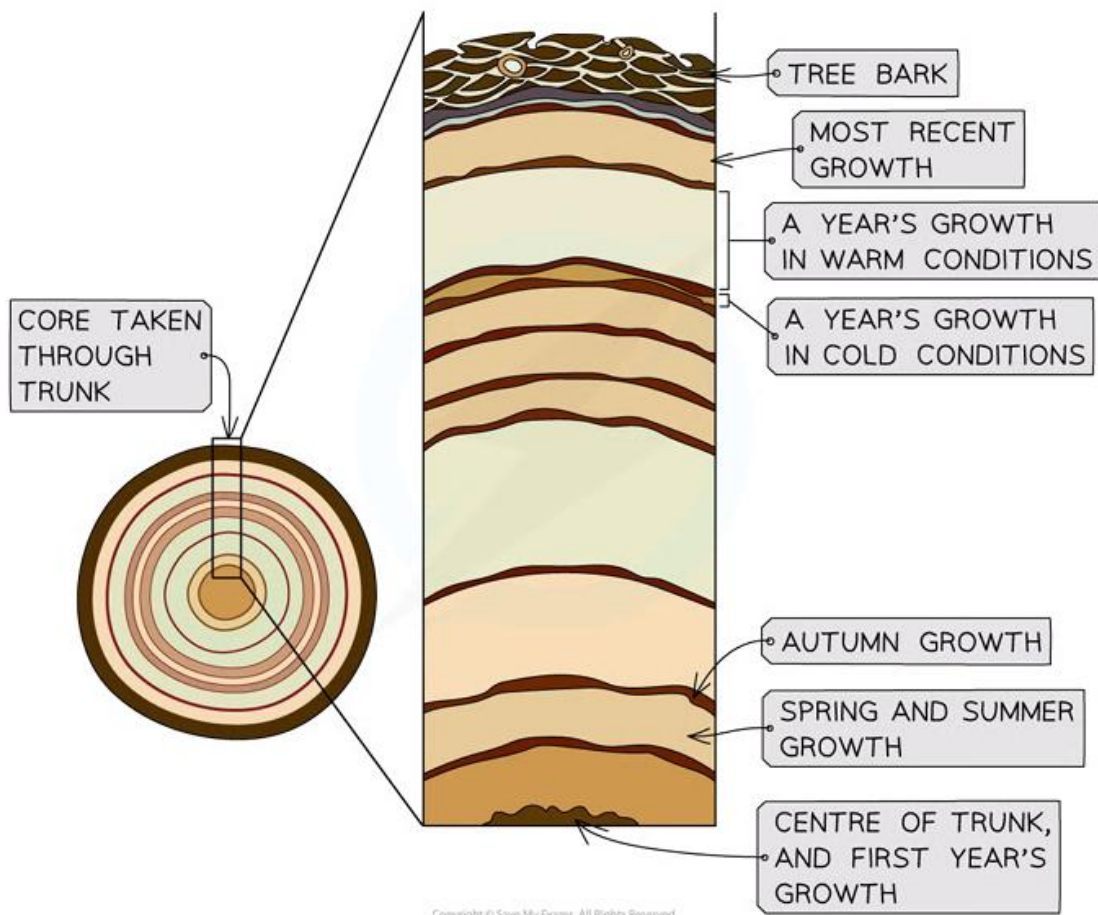
- Changes to the habitat have to take place before a different organism can survive there
- To make soil improvements
- This allows deeper soil bushes to grow



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2. Temperature Records

- Since 1850, temperature has been measured around the world using thermometers
- This gives reliable but short-term record of global temperature change
- Records from that age show overall trend in increasing average global temperatures
- Causes: Anthropogenic CO₂ Emission, Solar Flares, Volcanic Eruptions



3. Dendrochronology

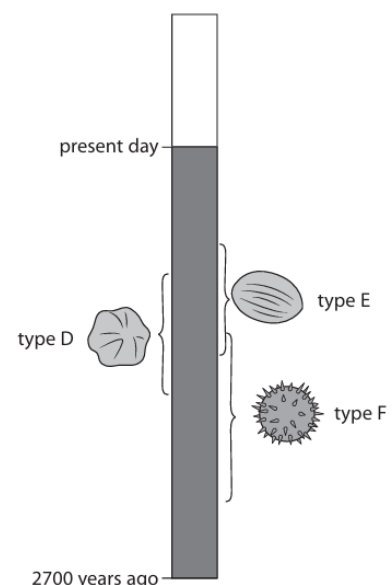
- Dendrochronology is the study of tree rings
- Every year a tree ring (xylem) is formed
- By counting the number of xylem rings, plant's age can be known
- Thickness of the ring is proportional to the temperature of the year in which the ring was formed
- As temperature increases → RUBISCO activity increases → Higher rate of photosynthesis → More GALP is produced → More biomass produced
- However, other factors such as light intensity, rainfall and CO₂ concentration can also contribute to thickness so it is not very accurate

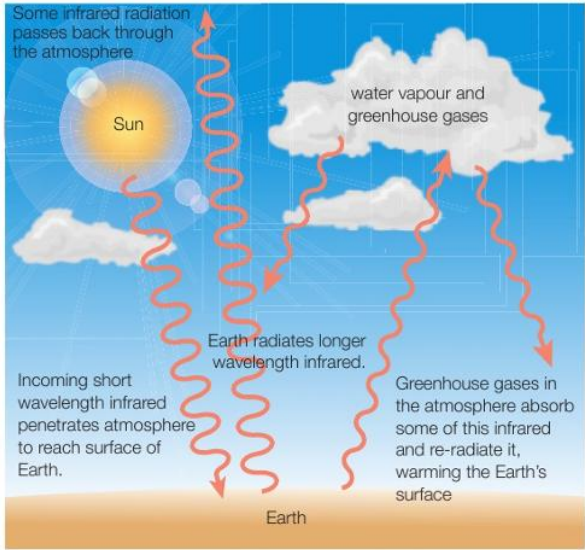
4. Peat Bogs

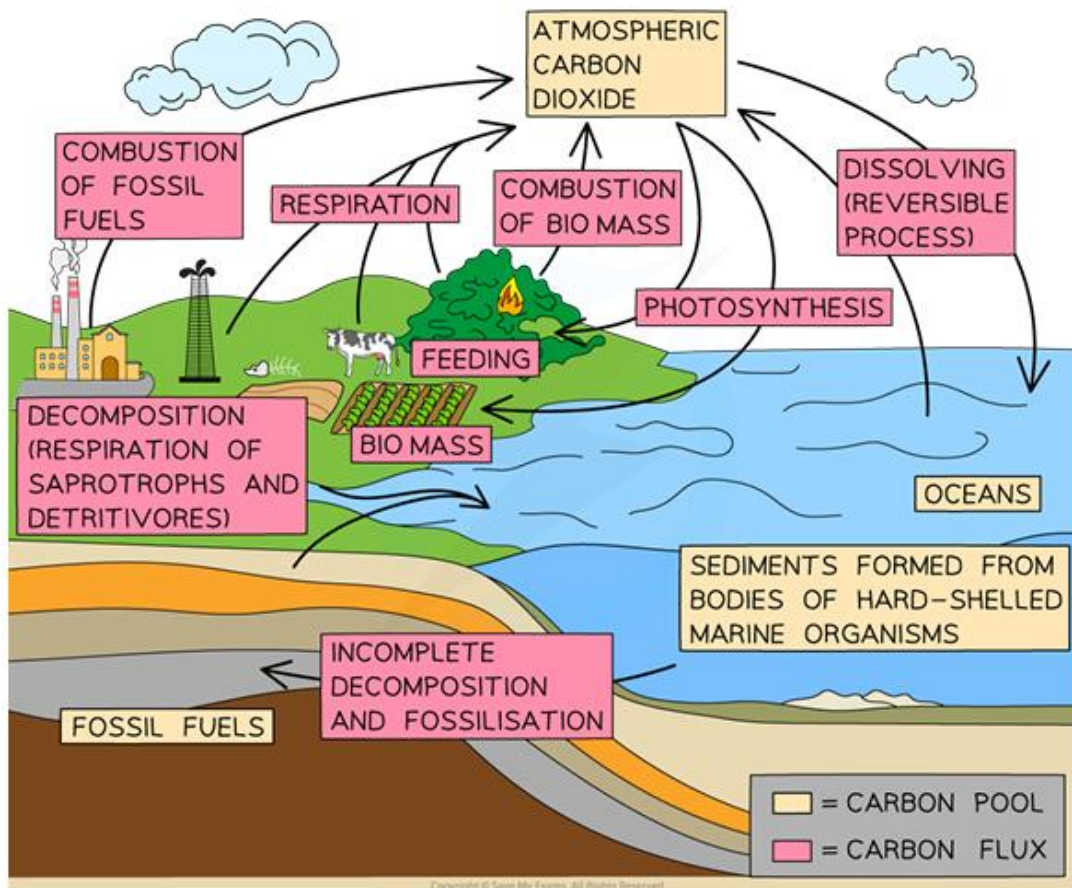
- Conditions in bog were not suitable for decomposition to take place
- **Cold temperatures** mean that microbial enzymes were working slowly
- **Low pH** (acidic conditions) is not suitable for bacteria to survive
- Water-logged bog meant that **oxygen levels** were very **low** (conditions are anaerobic)
- So not enough energy released by respiration
- Bog protected from insects and animals

→ Explain distribution of types of pollen grains in the peat bog / mud core:

- Each type of pollen grain of one species is present when conditions are favorable for that plant



	- Condition changes overtime, allowing different types of plants to grow - Previous type of plant disappears due to competition with another species or favorable conditions changed - Plants compete for light, space, water and nutrients
5.17	understand the causes of anthropogenic climate change, including the role of greenhouse gases in the greenhouse effect
	Anthropogenic Climate Change: change in mean temperature of an area over a long period of time caused by human activities Greenhouse Gases: - Carbon dioxide → Combustion, Aerobic respiration, Decomposition - Methane → Anaerobic respiration of bacteria in water logged paddy field, Stomach of ruminant animals → Decomposition in landfill sites - Water Vapor → Evaporation - CFC → Refrigerator Greenhouse Effect: 1. Incoming short wavelength passes through the greenhouse gases and reach surface of the Earth 2. The Earth surface radiates longer infrared wavelength. 3. Longer wavelength is absorbed by the greenhouses gases and they trap heat energy and reradiate back to the Earth, warming the surface → Without greenhouse effect, Earth would be at -63°C instead of 14°C. → It is the enhanced greenhouse effect that is causing the global warming.  The diagram illustrates the greenhouse effect. It shows the Sun emitting short wavelength infrared radiation (represented by blue wavy arrows) that passes through the atmosphere and reaches the Earth's surface. The Earth's surface then radiates longer wavelength infrared radiation (represented by red wavy arrows). Some of this radiation is absorbed by greenhouse gases in the atmosphere (water vapour and greenhouse gases), which then re-radiate it back towards the Earth's surface, warming it. Labels include: 'Sun', 'Some infrared radiation passes back through the atmosphere', 'water vapour and greenhouse gases', 'Earth radiates longer wavelength infrared.', 'Incoming short wavelength infrared penetrates atmosphere to reach surface of Earth.', and 'Greenhouse gases in the atmosphere absorb some of this infrared and re-radiate it, warming the Earth's surface'.
5.18	understand how knowledge of the carbon cycle can be applied to methods to reduce atmospheric levels of carbon dioxide



Storage of Carbon:

As Carbon Sink: a reservoir where carbon dioxide is removed from atmosphere and locked into organic and inorganic compounds

→ Humus, Rocks, Fossil Fuels, Inside the organism body as biomass, Oceans, Coral reefs, Limestone.

Absorption of Carbon:

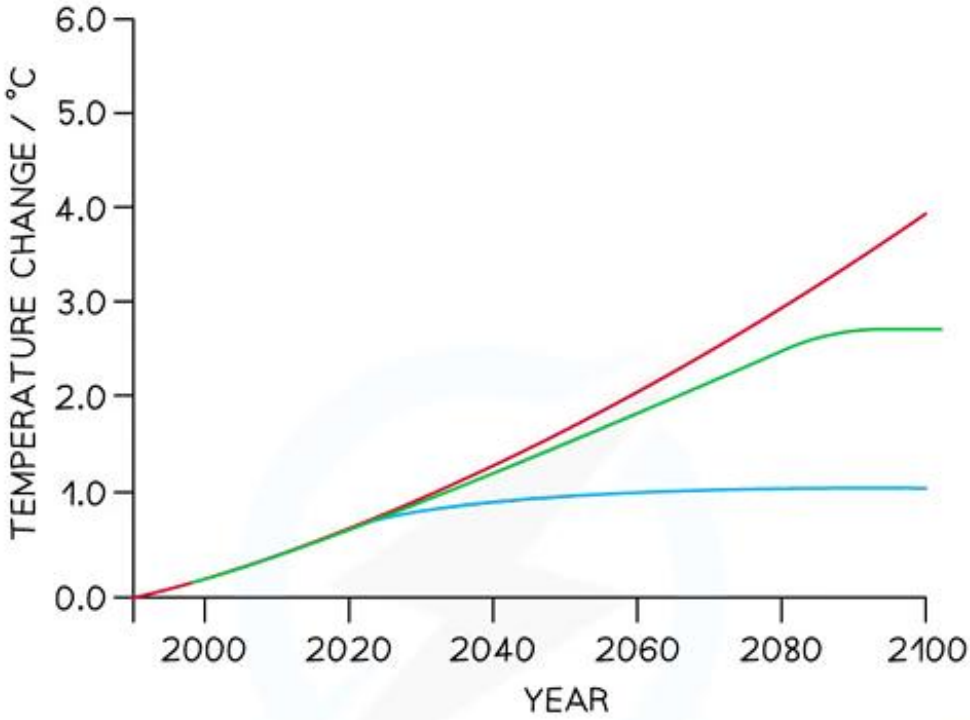
Photosynthesis: Light-independent reaction fixes carbon dioxide into GP

Emission of Carbon:

- **Aerobic Respiration**
- **Decomposition**
→ Break down organic matter with digestive enzymes
- **Combustion of Fossil Fuels**
- **Volcanic Eruptions**
- **Weathering**
- **Forest Fires**

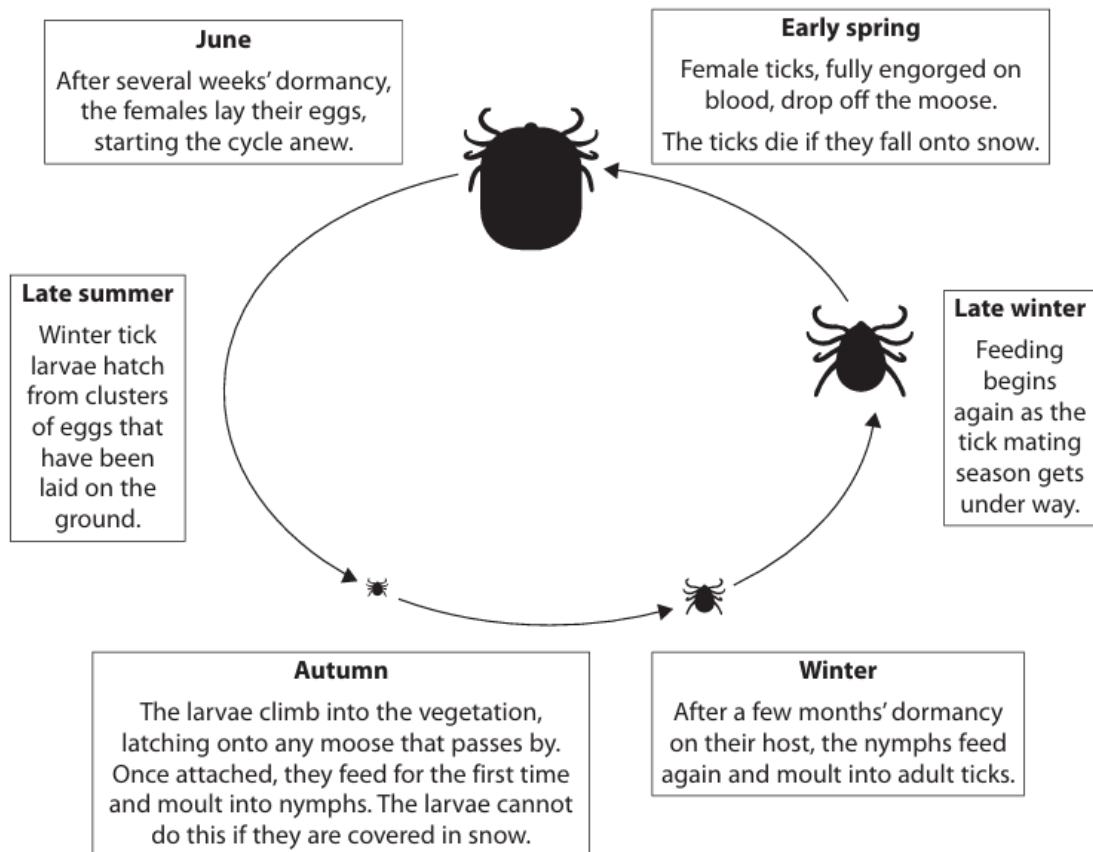
5.19

- (i) understand that data can be extrapolated to make predictions and that these are used in models of future climate change
- (ii) understand that models for climate change have limitations

i	Extrapolation: Use line of best fit from existing data relating to global warming to make predictions about global temperatures → This allows to build models to show how the climate may change in the future  The graph plots Temperature Change (°C) on the y-axis (0.0 to 6.0) against Year on the x-axis (2000 to 2100). Three lines represent different scenarios: a blue line that plateaus at ~1.1°C, a green line that peaks at ~2.7°C around 2090, and a red line that rises linearly to ~4.0°C by 2100. KEY: — = HUMANS MANAGE TO SIGNIFICANTLY REDUCE GREENHOUSE GAS EMISSIONS — = GREENHOUSE GAS EMISSIONS CONTINUE TO INCREASE FOR A WHILE BEFORE STARTING TO DECREASE — = GREENHOUSE GAS EMISSIONS CONTINUE TO INCREASE ON THEIR CURRENT TRAJECTORY Copyright © Save My Exams. All Rights Reserved
ii	Limitations of Prediction Models - Other new factors may affect climate the future - No one knows which scenario is most likely to occur - Future technologies might be successful at removing greenhouse gases from the atmosphere
5.20	understand the effects of climate change (changing rainfall patterns and changes in seasonal cycles) on plants and animals (distribution of species, development and lifecycles)
	Effect of Climate Change on Plants and Animals 1. Change rainfall patterns: Some areas rain a lot which might lead to flood which destroys crops and disrupt habitats; Some area might not rain at all which results in drought which also makes unable to grow crops and dehydration can occur 2. Changes in seasonal cycles: Organisms are adapted to the timing to the seasons to mate, to hibernate and to prey; Changing

- Both affect distribution of species and development of life cycles
- Both can result in migration of species to less hot and safer areas due to climate change.
- This might be due to the enzymes not working efficiently or their prey migrating to other places.
- This can also avoid dehydration.

*(ii) The diagram shows the life cycle of the tick.



The ticks feed on the blood of the moose. One tick can remove 200 to 600 times its body mass.

The ticks irritate the moose. The moose will scratch against trees causing large clumps of fur to fall off.

Explain how global warming might affect the life cycle of ticks and result in the decline in the number of moose.

Use all the information in the question to support your answer.

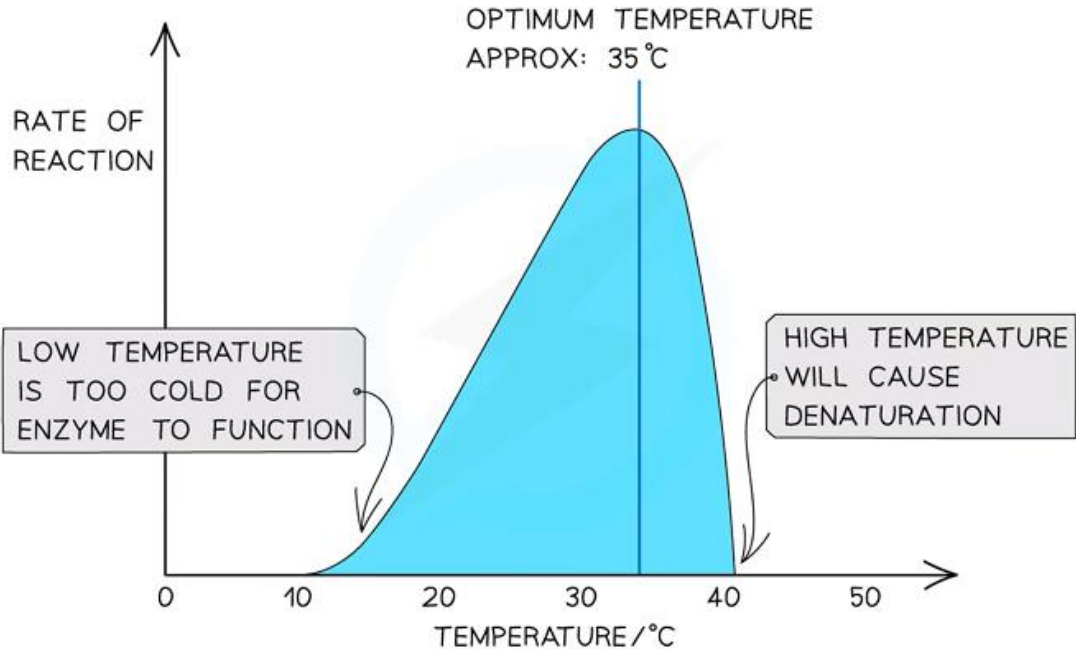
(6)

Global Warming:

- global warming will increase the temperature of the earth's {surface / atmosphere}
- winters will get warmer so less snow
- winters will get shorter so snow present for fewer days

Effect of Temperature on ticks:

- warmer conditions decrease life cycle time
- fewer ticks will die in the snow in early spring
- more females to lay eggs
- larvae less likely to be covered in snow in autumn
- so, more larvae become nymphs

	*Check the mark scheme for the full interpretation: the answers are extracted just for this specification point.
5.21	understand the effect of temperature on the rate of enzyme activity and its impact on plants, animals and microorganisms, to include Q ₁₀
	At high temperatures enzymes can denature  Effect of Temperature on Enzyme: - As temperature increases, rate of enzyme reaction increases - This is because more kinetic energy leads to more frequent collisions between enzymes and substrates - So, DNA synthesis, mitosis, protein synthesis, respiration would be faster, contributing to faster growth Q₁₀ Value - Temperature coefficient or Q₁₀ value for a reaction shows how much the rate of a reaction changes when the temperature is raised by 10° C - At temperatures before the optimum, a Q₁₀ value of 2 means the rate doubles when the temperature is raised by 10° C. $Q_{10} = \frac{\text{rate of enzyme activity at } T + 10^{\circ} \text{C}}{\text{rate of enzyme activity at } T^{\circ} \text{C}}$
5.22	CORE PRACTICAL 12 Investigate the effects of temperature on the development of organisms (such as seedling growth rate or brine shrimp hatch rates), considering the ethical use of organisms.
5.23	understand how evolution (a change in allele frequency) can come about through gene mutation and natural selection

	Evolutionary Mechanisms: causes differences in allele frequencies to accumulate 1. Mutation: changes in base sequence of the gene 2. Natural Selection: the process where individuals with advantageous traits are more likely to survive and reproduce, increasing the frequency of those alleles in a population 3. Genetic Drift: random change in allele frequencies in a population due to chance events, especially in small populations such as population bottleneck or Founder's effect
5.24	understand how isolation reduces gene flow between populations, leading to allopatric or sympatric speciation
	1 Population of individuals ● = individual organism 2 Physical barriers stop interbreeding and gene flow between populations. 3 Populations adapt to new environments. 4 Allele and phenotype frequency change leading to development of new species. Species: Group of living organisms that can reproduce to give fertile offspring Hybrid: Resulting offspring of two different species interbreeding (infertile) Subspecies: Varieties of the same species that can still reproduce to give fertile offspring despite having phenotypic or behavioral differences (2) Types of Speciation: 1. Allopatric Speciation: formation of new species due to geographical isolation that prevents gene flow between populations 2. Sympatric Speciation: formation of new species within the same geographical area due to reproductive isolation (e.g. ecological, behavioral, or temporal differences) Speciation requires (2) conditions: 1. Isolation Mechanism (Prevents Gene Flow) 2. Evolutionary Mechanism (Change Allele Frequencies) 1. Isolation Mechanism: a process that prevent gene flow (exchange of alleles) between populations. Types of Isolation Mechanism: 1. Geographical Isolation (Allopatric) 2. Ecological Isolation (Sympatric) 3. Seasonal Isolation (Sympatric) 4. Behavioral Isolation (Sympatric) 5. Mechanical and Gametic Isolation (Sympatric) 2. Evolutionary Mechanisms: causes differences in allele frequencies to accumulate 1. Mutation 2. Natural Selection 3. Genetic Drift Isolation Mechanism + Evolutionary Mechanism → Genetic Differences → Reproductive Isolation → Speciation

5.25	understand the way in which scientific conclusions about controversial issues, such as what actions should be taken to reduce climate change, or the degree to which humans are affecting climate change, can sometimes depend on who is reaching the conclusions
	Why some people consider anthropogenic climate change theory is controversial: - Just because there is a correlation between global temperature and increasing greenhouse gas emission, does not mean there is a causation - Certain politicians and business people have different opinions and want to continue their activities - Other activities such as volcanic eruptions and solar flares can cause climate change as well - Due to lack of education and lack of trust in scientists
5.26	understand how reforestation and the use of sustainable resources, including biofuels, are examples of the effective management of the conflict between human needs and conservation
	Sustainability: production of products that are carbon neutral and biodegradable so that the future generation do not need to compromise for that How can effects of anthropogenic climate change be reduced: 1.Reforestation: more plants will absorb more carbon dioxide for photosynthesis and tresses are carbon sinks 2.Biofuel/ Solar/Wind Power: reduce carbon dioxide released into the atmosphere 3.Reduce cattle being farmed: reduce the methane being released into the atmosphere 4.Use GM crops: more drought/insect resistant crops so grow higher yield and prevent starvation 5.Recycle Plant Material as Fertilizers: prevents eutrophication, bioaccumulation due to artificial fertilizers

Notice

- In this booklet, mark scheme points from past paper questions (mostly 2020-2023 and a few from 2010-2019) are included for each specification point to connect the syllabus and the past paper questions.
- For less frequently examined specification points, Pearson Edexcel IAL textbook and Save My Exams (available at [savemyexams.com](https://www.savemyexams.com)) notes have been the main reference of this booklet.
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